



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Subject & Code: Statistical Analysis (21AI62)

Semester: VI Semester

Faculty Name: Mr. K. S. Santhosh Kumar

Question Bank – Module 5

Sl. No. Questions

1. Define sampling theory and its importance in statistical analysis.
2. List the different methods of sampling.
3. Explain the principles of
 - i. Simple random sampling with replacement (SRSWR) and without replacement (SRWOR)
 - ii. Stratified random sampling
 - iii. Systematic sampling
4. Describe the central limit theorem and its significance in sampling theory.
5. Explain the concept of statistical significance.
6. Define tests of significance and their role in statistical analysis.
7. Briefly explain the types of tests of significance:
 - i. z-test
 - ii. t-test
 - iii. Chi-square test
 - iv. F-test
8. Calculate sample sizes using appropriate sampling techniques.
9. Explain the basic principles and assumptions of:
 - i. z-test
 - ii. t-test (including one-sample t-test, independent samples t-test, paired samples t-test)
 - iii. Chi-square test (including chi-square test of independence and chi-square goodness of fit test)
 - iv. F-test (including one-way ANOVA)
10. Analyze the impact of sample size on the accuracy of statistical tests.
11. Critically assess the assumptions underlying different tests of significance.
12. Compare different types of sampling techniques and their advantages/disadvantages.
13. Evaluate the reliability of hypothesis tests based on sample data.
14. Critically assess the assumptions underlying different tests of significance.
15. Design a sampling plan for a research study based on specific criteria and objectives.
16. Develop a hypothesis test scenario based on real-world data and interpret the results.
17. Solve problems related to sampling theory and tests of significance:
 - i. Calculate sample sizes for different sampling methods.
 - ii. Conduct hypothesis tests (z-test, t-test, chi-square test, F-test) for specific research questions.
 - iii. Evaluate and interpret the results of hypothesis tests, drawing conclusions based on statistical significance.